

Assignment 4

1. Main idea of diffusion models

Diffusion models generate data by learning how to reverse a gradual noising process. The core idea is:

slowly destroy an image with noise, then train a neural network to reconstruct it step by step.

In the forward process, A real image is gradually corrupted by adding small amounts of Gaussian noise over many steps. until it becomes pure random .

In the reverse process, the model learns how to remove this noise step by step to reconstruct a clean image.

2. Why add noise during training?

Noise is added during training so the model can learn how images become corrupted and how to reverse that corruption. By seeing noisy images at many different noise levels, the network learns to remove noise gradually and reconstruct meaningful structure.

The training process teaches the model:

- how noise affects images
- how to recover patterns from noisy inputs
- how to denoise step by step

3. Why predict noise instead of the clean image?

1. because the noise pattern has a simpler and more consistent structure than full images.
2. The model only needs to estimate what noise was added rather than reconstructing every image detail directly.
3. This makes the loss function smoother and easier to optimize.
As a result, training becomes more stable and converges better.

4. Compare GANs and diffusion models

GANs use two networks: a generator and a discriminator competing against each other using a minimax game

Training is unstable because if one network becomes stronger, the other may fail, causing problems like mode collapse.

Diffusion models are more stable because they do not use adversarial training. Instead, they train a single network to predict noise using supervised regression, so optimization is more stable and easier to train.

Diffusion models usually produce more stable and diverse results.

5. Why is high-resolution diffusion expensive?

High-resolution images contain millions of pixels, so applying diffusion directly requires huge memory and computation.

The model must process many denoising steps for every pixel, which is very slow.

Latent diffusion solves this by first compressing the image into a smaller latent representation using a VAE.

Diffusion is then applied in the latent space instead of pixel space, greatly reducing computation while preserving image quality.

The process is:

1. Compress image using a VAE encoder
2. Run diffusion on the smaller latent representation
3. Decode the latent back into an image

6. Why add a GAN-style discriminator to the VAE decoder?

A normal VAE often produces blurry images because reconstruction losses average pixel values.

Adding a GAN-style discriminator encourages the decoder to generate sharper and more realistic textures.

The discriminator checks whether images look real, forcing the decoder to improve visual quality.

This helps solve the blur problem in VAEs.

From Diffusion.ipynb

[Assignment 4](#)

1. Why is noise prediction supervised learning?

During training, we already know:

- the original image
- the added noise
- the noisy image

So the model receives an input (noisy image) and a correct target output (the noise). This makes it a supervised learning problem with a clear objective.

Unlike GANs, there is no adversarial competition between networks, so training is more stable and easier to optimize.

2. What happens during visualization and sampling?

As noise increases, the image gradually loses details and structure until it becomes almost pure random noise.

During sampling, the reverse happens: the noisy image slowly becomes clearer and more

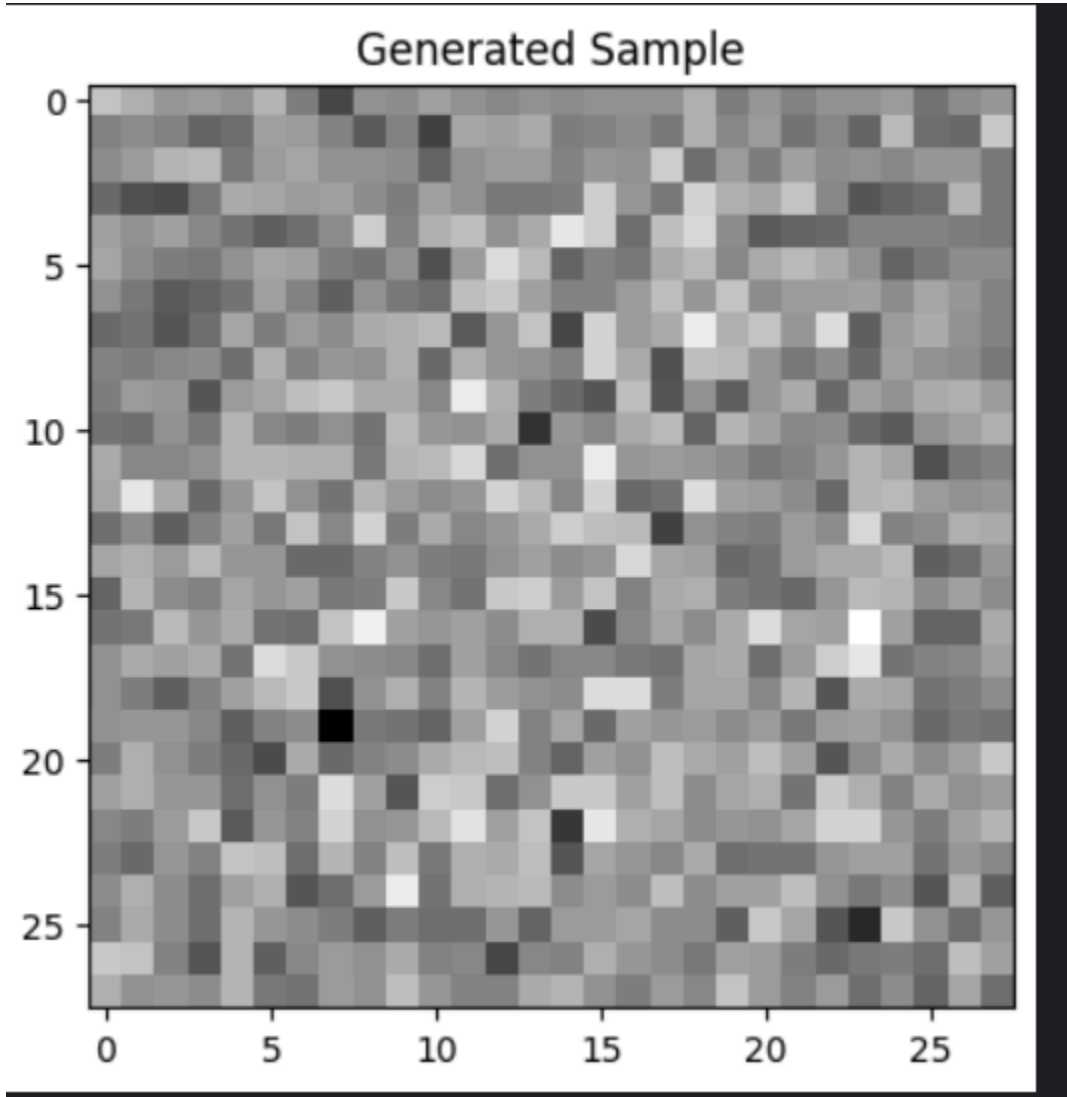
structured.

Shapes and patterns appear step by step until a final image is generated.

This demonstrates the main idea of diffusion models:

- forward process = destroy information with noise
- reverse process = recover information by removing noise

3. Describe the generated image



The generated image may contain rough shapes or patterns similar to the training data, but it is usually blurry or unrealistic.

This happens because the model was trained for only a small number of steps and on a simple dataset.

The network architecture is also relatively small, so it cannot capture complex image details well.

Better realism usually requires larger datasets, stronger models, and longer training.

at is why diffusion sampling can take much longer than autoregressive generation.